

Day – 10 DHCP Configuration Theory.

Task – 1 Config DHCP server.

- Step 1: Build the Topology
- Open Cisco Packet Tracer and drop the following devices onto your workspace:
- 1 Switch (e.g., Model 2960).
- 1 Server (Labeled as Server-PT, representing your DHCP server).
- 2 PCs (Labeled as PC0 and PC1).
- Use Copper Straight-Through cables to connect the Server and both PCs to individual ports on the Switch.
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- Step 2: Configure the DHCP Server
- Before handing out addresses to clients, the DHCP server itself needs a static IP address:
- Click on the Server, go to the Desktop tab, and select IP Configuration.
- Assign a static IP configuration:
- IPv4 Address: 192.168.1.254
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Next, navigate to the Services tab at the top of the server window and select DHCP from the left-hand menu.
- Configure your DHCP pool settings:
- Ensure the service status is set to ON.
- Pool Name: HomePool (or leave the default).
- Default Gateway: 192.168.1.1
- DNS Server: 8.8.8.8
- Start IP Address: 192.168.1.10 (this ensures the pool starts handing addresses from .10 upward).
- Subnet Mask: 255.255.255.0
- Click the Save (or Add) button to save your pool configuration.
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- Step 3: Configure the Client PCs
- Click on PC0, go to the Desktop tab, and open IP Configuration.
- Instead of *Static*, select the DHCP radio button. Watch closely as it sends a broadcast request and successfully populates with an IP address assigned from your server pool.
- Repeat the exact same process for PC1.
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- DHCPACK (Acknowledgment): The server finalizes the transaction, officially registering the lease to the client's MAC address and delivering the final configuration parameters (Subnet Mask, Default Gateway, DNS servers).

Task – 3 Change the DHCP range.

- In a real Cisco Packet Tracer lab, configuring a DHCP pool with only a single available IP address (for example, setting both the start and maximum number of users to just 1) creates an immediate address exhaustion scenario.
- What Happens in the Simulation:
- PC0 (First Client): Successfully completes the DORA handshake and is assigned the single available IP address from the pool.
- PC1 (Second Client): Broadcasts its DHCPDISCOVER request, but because no free leases remain in the pool, it receives no valid allocation and falls back to an APIPA (Automatic Private IP Addressing) address in the 169.254.x.x range.
- Why Only One PC Succeeds:
- Only one PC succeeds because the DHCP server's address pool is completely exhausted after issuing its only available lease. Without free IP addresses remaining to allocate, any subsequent client requests are denied or left unanswered, forcing the extra device to self-assign a local non-routable address.

Task – 4 Wifi scenario in cisco packet tracer.

- Simulating the Zomato Restaurant Wi-Fi Scenario
- To set this up in Packet Tracer:
- The Infrastructure: Place a Wireless Router (or a Switch connected to a Wireless Access Point), a Server configured as the restaurant's local DHCP server, a Laptop, and a Smartphone (you can use a Generic PC or a smartphone icon if available in your version).
- The DHCP Scope: Configure the DHCP server pool with a scope designed for customer traffic, such as:
- Network Subnet: 192.168.10.0 /24
- Default Gateway: 192.168.10.1
- Pool Range: 192.168.10.10 to 192.168.10.100
- Simulated Results & Assigned IPs
- When the devices connect to the restaurant's wireless network and request an address via the DORA process, they receive dynamic leases:

- Restaurant Laptop (PC0): 192.168.10.10 (Subnet: 255.255.255.0, Gateway: 192.168.10.1)
- Customer Smartphone (PC1): 192.168.10.11 (Subnet: 255.255.255.0, Gateway: 192.168.10.1)
- Why DHCP is Crucial in a Restaurant Scenario
- High Device Turnover (Mobility): Customers arrive, order food, browse Zomato for 45 minutes, and leave. DHCP automatically reclaims expired IP leases (via lease time configurations) and recycles them for new incoming patrons.
- Zero Configuration for Customers: Diners don't want to manually type in IP addresses, subnet masks, and DNS servers just to connect to the Wi-Fi. DHCP handles the entire handoff invisibly in the background.
- Prevents IP Conflicts: With dozens of devices connecting simultaneously, manual IP assignment would inevitably lead to two phones getting the same address, causing connection drops. DHCP ensures every active device gets a strictly unique identifier.
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